The litetable Package — Colorful Timetable[*†]

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1 Introduction

The litetable package provides a colorful design of timetable, developed by expl3 based on Ti*k*Z. It supports various compilation methods, such as pdfLaTeX, XeLaTeX, ApLaTeX, LuaLaTeX, etc. Click to jump to the manual's [Chinese Version] [Cantonese Version].

2 Usage

To load this package, write the line

```
\usepackage [scale = ⟨fp num⟩ ] {litetable}
```

`litetable` (*env.*) The `scale` key can set the scale of the timetable (Default: `1.0`). The `litetable` environment can create a blank timetable frame, and it should be executed after commands `\timelist` and `\weeklist`.

```
\begin{litetable} [⟨keys⟩] {⟨title⟩} [⟨keys⟩] ... \end{litetable}
```

The mandatory argument can set the title of the timetable, and the optional argument accepts the following keys

`color` = ⟨*string*⟩ can set the background color (Default: `gray`), this key's name could be omitted.

`sem` = ⟨*string*⟩ can set the semester information at the northeast corner of the page.

`hline` = ⟨*string*⟩ can set style of the horizontal lines (Default: `solid`).

`\weeklist` `\weeklist [⟨keys⟩] {⟨list⟩} [⟨keys⟩]`

The mandatory argument accepts an array to set a list of working days and the width of each column at the top of the timetable. The optional argument accepts the following keys

`format` = ⟨*format commands*⟩ can set the font for the list of working days (Default: `\bfseries`).

`sep` = ⟨*dim*⟩ can set the separator of the list of working days.

```
\weeklist [ format = \bfseries \scshape, sep = \textbar ]
  { Mon -> 1.05, Tue -> 1.05, Wed -> 1.1, Thu -> 1.1, Fri -> .9 }
```

[*] https://github.com/myhsia/litetable, https://ctan.org/pkg/litetable

[†] Lijun Guo developed an interface to parse ⟨*left*⟩ -> ⟨*right*⟩ data structures.

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`\timelist` `\timelist [⟨keys⟩] {⟨list⟩} [⟨keys⟩]`

The mandatory argument accepts an array to set the time list on the left side of the timetable. The optional argument accepts the following keys

numformat = ⟨*format commands*⟩ can set the font for the sequence number of the time list (Default: `\ttfamily \bfseries`).

timeformat = ⟨*format commands*⟩ can set the font for the time of the time list. (Default: `\ttfamily`).

hidetime = ⟨*true* | **false**⟩ hide the time in the time list and only retain the sequence number. The initial value is `false`.

```
\timelist [ numformat = \bfseries, timeformat = \ttfamily ]
  { 08:30 -> 10:00, 10:30 -> 12:00, 13:00 -> 14:30, 15:00 -> 16:30 }
```

`\course` `\course [⟨keys⟩] {⟨start⟩} [⟨keys⟩] {⟨end⟩} [⟨keys⟩]`

It's used to add course boxes on the current workday, and needs to be executed within the `litetable` environment. The two mandatory arguments can set the start and ends of the course respectively, the optional argument accepts the following keys

color = ⟨*string*⟩ can set the color of the course box (Default: `teal`), this key's name could be omitted.

subject = ⟨*string*⟩ can set the name of the course.

location = ⟨*string*⟩ can set the location of the course.

lecture = ⟨*string*⟩ can set the lecture of the course.

comment = ⟨*string*⟩ can add footnote to the course.

TEXhackers note:

- If ⟨*start*⟩ = ⟨*end*⟩ (the height of the course box is 1 unit), then **location** and **lecture** will be outputted in the same line and **comment** will be hidden.
- The template will correct automatically if ⟨*start*⟩ and ⟨*end*⟩ were misplaced.
- If neither **location** nor **lecture** is assigned value, then **subject** will be outputted in the vertical center of the course box.
- Course boxes that exceed the range of the timetable won't display and it will return a warning. The input example refers to Appendix 3.

`\newday` `\newday [⟨integral value⟩]`

It can move the next course boxes right ⟨*integral value*⟩ working days. The default value of the optional argument is 1.

`\more` `\more {⟨comment⟩}`

It can add a comment at the southwest corner of the timetable.

3 Working Example

```
\documentclass[svgnames, a4paper]{article}

\usepackage[scale = .9]{litetable}
\usepackage{twemojis}
\usepackage[osf, mono = false]{libertine}
\usepackage[T1]{fontenc}

\begin{document}

\weeklist [ format = \bfseries \scshape, sep = \textbar ]
  {
    \texttwemoji{1f312} Mon -> 1.05, \texttwemoji{1f525} Tue -> 1.05,
    \texttwemoji{1f30a} Wed -> 1.1,  \texttwemoji{1f332} Thu -> .9,
    \texttwemoji{1fa99} Fri -> .9
  }
\timelist [ numformat = \ttfamily \bfseries, timeformat = \ttfamily ]
  {
    08:05 -> 08:50, 08:55 -> 09:40, 10:00 -> 10:45, 10:50 -> 11:35,
    11:40 -> 12:25, 13:30 -> 14:15, 14:20 -> 15:05, 15:15 -> 16:00,
    16:05 -> 16:50, 18:30 -> 19:15, 19:20 -> 20:05, 20:10 -> 20:55
  }
\begin{litetable} [ DarkBlue, sem = SEM 7, hline = dashed ]
  {Course Schedule}
  \course [ subject = interface3, comment = \TeX~Live 2025,
            lecture = The \LaTeX\ Project, DarkBlue ] {4} {5}
  \newday
  \course [ subject = expl3, lecture = The \LaTeX\ Project ] {8} {8}
  \newday
  \course [ subject = Keep on \TeX ing, lecture = Donald E. Knuth,
            location = Stanford University, Purple ] {10} {11}
  \newday
  \course [ subject = Ti\textit k\/Z, lecture = \textsc{pgf},
            Crimson, comment = Version 3.1.10 ] {3} {5}
  \more {Programme Duration: 09 / 2021 -- 07 / 2025}
\end{litetable}

\end{document}
```

Course Schedule

SEM 7

	Mon	Tue	Wed	Thu	Fri
1 08:05 08:50					
2 08:55 09:40					
3 10:00 10:45				TikZ PGF Version 3.1.10	
4 10:50 11:35	interface3 The LaTeX Project TeX Live 2025				
5 11:40 12:25					
6 13:30 14:15					
7 14:20 15:05					
8 15:15 16:00		expl3 The LaTeX Project			
9 16:05 16:50					
10 18:30 19:15			Keep on TeXing Stanford University Donald E. Knuth		
11 19:20 20:05					
12 20:10 20:55					

Programme Duration: 09 / 2021 – 07 / 2025

A The Source Code

Start the optionlist package for l3docstrip.

```
⟨*package⟩
⟨@@=ltbl⟩
\ProvidesExplPackage {litetable} {2026-02-07} {v3.9A}
  {A Colorful Timetable Design}
\RequirePackage{tikz}
```

Warning Broadcast

```
\cs_new_protected:Npn \__ltbl_msg_new:nn #1#2
  { \msg_new:nnn { litetable } {#1} {#2} }
\cs_new_protected:Npn \__ltbl_msg_warning:n #1
  { \msg_warning:nn { litetable } {#1} }
\__ltbl_msg_new:nn { course }
  { \exp_not:N \course ~ box(s) ~ exceed ~ workdays ~ were ~ ignored }
```

`\__ltbl_get_left:nN` `\__ltbl_get_left:eN` `\__ltbl_get_right:nN` `\__ltbl_get_right:eN`

Handle ⟨*left*⟩ -> ⟨*right*⟩ data structures (by Lijun Guo)

```
\cs_new_protected_nopar:Npn \__ltbl_get_left:nN #1#2
  {
    \group_begin: \seq_set_split:Nnn \l__ltbl_tmpa_seq { -> } {#1}
    \exp_args:NNNe \group_end:
    \tl_set:Nn #2 { \seq_item:Nn \l__ltbl_tmpa_seq { 1 } }
  }
\cs_new_protected_nopar:Npn \__ltbl_get_right:nN #1#2
  {
    \group_begin: \seq_set_split:Nnn \l__ltbl_tmpa_seq { -> } {#1}
    \exp_args:NNNe \group_end:
    \tl_set:Nn #2 { \seq_item:Nn \l__ltbl_tmpa_seq { 2 } }
  }
\cs_generate_variant:Nn \__ltbl_get_left:nN  { eN }
\cs_generate_variant:Nn \__ltbl_get_right:nN { eN }
```

(*End of definition for* `\__ltbl_get_left:nN` *and* `\__ltbl_get_right:nN`.)

A.1 User's Interface

`\l__ltbl_textwidth_dim` `\l__ltbl_textheight_dim` `\l__ltbl_reswidth_dim` `\l__ltbl_resheight_dim`

Dimensions for the physical width and height of the timetable and margin, controlled by `\l_@@_scale_fp` in package option.

```
\dim_new:N \l__ltbl_textwidth_dim
\dim_new:N \l__ltbl_textheight_dim
\dim_new:N \l__ltbl_reswidth_dim
\dim_new:N \l__ltbl_resheight_dim
\keys_define:nn { litetable / pkgoption }
  {
    scale   .fp_set:N  = \l__ltbl_scale_fp,
    scale   .initial:n = 1.0,
    unknown .code:n    = \__ltbl_unknown_option:n {#1}
  }
```

(*End of definition for* `\l__ltbl_textwidth_dim` *and others.*)

`\__ltbl_unknown_option:n` Handel the unknown options.

\g__ltbl_base_options_clist

```
\clist_new:N \g__ltbl_base_options_clist

\cs_new_protected_nopar:Npn \__ltbl_unknown_option:n #1
  {
    \tl_if_empty:nTF {#1}
      {
        \clist_gput_right:NV
        \g__ltbl_base_options_clist \l_keys_key_str
      }
      {
        \exp_args:NNx
        \clist_gput_right:Nn \g__ltbl_base_options_clist
          { \l_keys_key_str = \exp_not:n {#1} }
      }
  }
```

(*End of definition for* `\__ltbl_unknown_option:n` *and* `\g__ltbl_base_options_clist`.)

Process the package option.

```
\ProcessKeyOptions [ litetable / pkgoption ]
\dim_gset:Nn \l__ltbl_textwidth_dim
  { \fp_eval:n { \l__ltbl_scale_fp     } \paperwidth    }
\dim_gset:Nn \l__ltbl_textheight_dim
  { \fp_eval:n { \l__ltbl_scale_fp     } \paperheight   }
\dim_gset:Nn \l__ltbl_reswidth_dim
  { \fp_eval:n { 1 - \l__ltbl_scale_fp } \paperwidth /2 }
\dim_gset:Nn \l__ltbl_resheight_dim
  { \fp_eval:n { 1 - \l__ltbl_scale_fp } \paperheight/2 }
```

\weeklist

Set a list of working days and the width of each column at the top of the timetable.

```
\NewDocumentCommand \weeklist { O{} m O{} }
  {
    \keys_set:nn { litetable / weeklist } { #1, #3 }
    \__ltbl_weeklist:n {#2}
  }
```

Key–value definitions for the `\weeklist` command.

\l__ltbl_weeklist_format_tl
\l__ltbl_weeklist_sep_tl

```
\keys_define:nn { litetable / weeklist }
  {
    format    .tl_set:N  = \l__ltbl_weeklist_format_tl,
      format  .initial:n = \bfseries,
    sep       .tl_set:N  = \l__ltbl_weeklist_sep_tl
  }
```

(*End of definition for* `\weeklist`, `\l__ltbl_weeklist_format_tl`, *and* `\l__ltbl_weeklist_sep_tl`. *This function is documented on page* 1.)

\timelist

Set the time list on the left side of the timetable.

```
\NewDocumentCommand \timelist { O{} m O{} }
  {
    \keys_set:nn { litetable / timelist } { #1, #3 }
    \__ltbl_timelist:n {#2}
  }
```

Key–value definitions for the `\timelist` command.

\l__ltbl_timelist_numformat_tl
\l__ltbl_timelist_timeformat_tl
\l__ltbl_timelist_hidetime_bool

```
\keys_define:nn { litetable / timelist }
  {
    numformat    .tl_set:N   = \l__ltbl_timelist_numformat_tl,
      numformat  .initial:n  = \ttfamily \bfseries,
    timeformat   .tl_set:N   = \l__ltbl_timelist_timeformat_tl,
      timeformat .initial:n  = \ttfamily,
    hidetime     .bool_set:N = \l__ltbl_timelist_hidetime_bool,
      hidetime   .initial:n  = false,
      hidetime   .default:n  = true
  }
```

(*End of definition for* `\timelist` *and others. This function is documented on page* 2.)

`litetable` (*env.*) Create a blank timetable frame.

```
\NewDocumentEnvironment { litetable } { O{} m O{} }
  {
    \clearpage \thispagestyle{empty}
    \group_begin:
    \keys_set:nn { litetable / frame } { #1, #3 }
    \tikzpicture [ remember ~ picture, overlay ]
      \__ltbl_maketable:n {#2}
  } { \endtikzpicture \group_end: \clearpage }
```

\l__ltbl_bg_color_tl
\l__ltbl_hline_type_tl
\l__ltbl_bg_sem_tl

Key–value definitions for the `litetable` command.

```
\keys_define:nn { litetable / frame }
  {
    color   .tl_set:N  = \l__ltbl_bg_color_tl,
      color .initial:n = gray,
    hline   .tl_set:N  = \l__ltbl_hline_type_tl,
      hline .initial:n = solid,
    sem     .tl_set:N  = \l__ltbl_bg_sem_tl,
    unknown .code:n    = \tl_if_novalue:nF {#1}
      { \tl_set_eq:NN \l__ltbl_bg_color_tl \l_keys_key_tl }
  }
```

(*End of definition for* `\l__ltbl_bg_color_tl`, `\l__ltbl_hline_type_tl`, *and* `\l__ltbl_bg_sem_tl`.)

`\course` Add course boxes on the current workday

```
\NewDocumentCommand \course { O{} m O{} m O{} }
  {
    \group_begin:
    \bool_lazy_any:nTF
      {
        {
          \int_compare_p:nNn { \l__ltbl_weekday_int } >
            { \clist_count:N \l__ltbl_week_clist }
        }
        { \int_compare_p:nNn {#2} > { \clist_count:N \l__ltbl_time_clist } }
        { \int_compare_p:nNn {#4} > { \clist_count:N \l__ltbl_time_clist } }
      } { \__ltbl_msg_warning:n { course } }
      {
        \keys_set:nn { litetable / course } { #1, #3, #5 }
        \int_compare:nNnTF {#2} < {#4}
```

```
          { \__ltbl_course_box_aux:nn {#2} {#4} }
          { \__ltbl_course_box_aux:nn {#4} {#2} }
      }
    \group_end:
  }
```

Key–value definitions for the `\course` command.

```
\keys_define:nn { litetable / course }
  {
    color    .tl_set:N  = \l__ltbl_course_color_tl,
      color  .initial:n = black,
    subject  .tl_set:N  = \l__ltbl_course_subject_tl,
    lecture  .tl_set:N  = \l__ltbl_course_lecture_tl,
    location .tl_set:N  = \l__ltbl_course_location_tl,
    comment  .tl_set:N  = \l__ltbl_course_comment_tl,
    unknown  .code:n    = \tl_if_novalue:nF {#1}
      { \tl_set_eq:NN \l__ltbl_course_color_tl \l_keys_key_tl }
  }
```

(*End of definition for* `\course` *and others. This function is documented on page* 2.)

`\more` Add a comment at the southwest corner of the timetable.

```
\NewDocumentCommand \more { m }
  {
    \node [ yshift = .5\l__ltbl_time_vunit_dim, left = 1ex,
            darkgray, font = \small \bfseries
          ] at ([shift =
              {(-\l__ltbl_reswidth_dim, \l__ltbl_resheight_dim)}
                ]current ~ page.south ~ east) {#1};
  }
```

(*End of definition for* `\more`*. This function is documented on page* 2.)

`\newday` Move the next course boxes right ⟨*integral value*⟩ working days.

```
\NewDocumentCommand \newday { O{1} } { \int_add:Nn \l__ltbl_weekday_int {#1} }
\int_new:N \l__ltbl_weekday_int
\int_set:Nn \l__ltbl_weekday_int { 1 }
```

(*End of definition for* `\newday`*. This function is documented on page* 2.)

A.2 Internal Auxiliary

`\l__ltbl_week_ratio_clist` `\l__ltbl_week_accum_clist` `\l__ltbl_week_num_fp` `\l__ltbl_week_hunit_dim`

The ratios of every working days, the accumulation of the ratios of every working days, the sequence number of every working days, the horizontal width unit of the timetable.

```
\clist_new:N \l__ltbl_week_ratio_clist
\clist_new:N \l__ltbl_week_accum_clist
\fp_new:N \l__ltbl_week_num_fp
\dim_new:N \l__ltbl_week_hunit_dim
```

(*End of definition for* `\l__ltbl_week_ratio_clist` *and others.*)

__ltbl_weeklist:n Define the auxiliary command of \weeklist.

```
\cs_new_protected_nopar:Npn \__ltbl_weeklist:n #1
  {
    \clist_set:Nn \l__ltbl_week_clist {#1}
    \exp_args:NNf \clist_map_inline:Nn \l__ltbl_week_clist
      {
        \__ltbl_get_right:eN {##1} \l__ltbl_tmpb_tl
        \clist_put_right:Ne \l__ltbl_week_ratio_clist { \l__ltbl_tmpb_tl }
      }
    \int_step_inline:nn { \clist_count:N \l__ltbl_week_clist }
      {
        \clist_clear:N \l__ltbl_week_accumtmp_clist
        \int_step_inline:nn {##1}
          {
            \clist_put_right:Ne \l__ltbl_week_accumtmp_clist
              { \clist_item:Nn \l__ltbl_week_ratio_clist {####1} }
          }
        \clist_put_right:Ne \l__ltbl_week_accum_clist
          { \fp_eval:n { \clist_use:Nn \l__ltbl_week_accumtmp_clist { + } } }
      }
    \fp_set:Nn \l__ltbl_week_num_fp
      {
        \clist_item:Nn \l__ltbl_week_accum_clist
          { \clist_count:N \l__ltbl_week_clist }
      }
    \dim_set:Nn \l__ltbl_week_hunit_dim
      { \fp_eval:n { 14/\l__ltbl_week_num_fp/15 } \l__ltbl_textwidth_dim }
  }
```

(*End of definition for* __ltbl_weeklist:n.)

\l__ltbl_time_num_int
\l__ltbl_time_vunit_dim
The sequence number of the time list, and the vertical gap between the start and end time.

```
\int_new:N \l__ltbl_time_num_int
\dim_new:N \l__ltbl_time_vunit_dim
```

(*End of definition for* \l__ltbl_time_num_int *and* \l__ltbl_time_vunit_dim.)

__ltbl_timelist:n Define the auxiliary command of \timelist.

```
\cs_new_protected_nopar:Npn \__ltbl_timelist:n #1
  {
    \clist_set:Nn \l__ltbl_time_clist {#1}
    \int_set:Nn \l__ltbl_time_num_int { \clist_count:N \l__ltbl_time_clist }
    \dim_set:Nn \l__ltbl_time_vunit_dim
      { \fp_eval:n { 1/(2\l__ltbl_time_num_int + 3.5) } \l__ltbl_textheight_dim }
  }
```

(*End of definition for* __ltbl_timelist:n.)

\l__ltbl_timelist_shift_dim Store the vertical shift of the sequence number of the time list.

```
\dim_new:N \l__ltbl_timelist_shift_dim
```

(*End of definition for* \l__ltbl_timelist_shift_dim.)

`\__ltbl_maketable:n` Define the auxiliary command of the `litetable` environment.

```
\cs_new_protected_nopar:Npn \__ltbl_maketable:n #1
  {
    \fill [ \l__ltbl_bg_color_tl!5 ]
      ([shift =
        {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
       ]current ~ page.north ~ west) rectangle +
      (\l__ltbl_textwidth_dim, -1.5\l__ltbl_time_vunit_dim)
     node [ midway, black, font = \huge \bfseries ] {#1};
    \tl_if_empty:NF \l__ltbl_bg_sem_tl
      {
        \node [ shift = {(-.02\paperwidth, -.75\l__ltbl_time_vunit_dim)},
                left, rectangle, fill = DarkBlue!10, text = DarkBlue!60,
                inner ~ sep = 2ex, rounded ~ corners = 8pt, font = \large
              ] at ([shift =
                  {(-\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                    ]current ~ page.north ~ east) { \l__ltbl_bg_sem_tl };
      }
    \int_step_inline:nnnn { 0 } { 2 } { \l__ltbl_time_num_int }
      {
        \bool_lazy_and:nnTF
          { \int_if_even_p:n { \l__ltbl_time_num_int }             }
          { \int_compare_p:nNn { ##1 } = { \l__ltbl_time_num_int } }
          {
            \filldraw [ fill = \l__ltbl_bg_color_tl!5, thick,
                        draw = gray, \l__ltbl_hline_type_tl ]
              ([shift = {(
                {-.4pt + \l__ltbl_reswidth_dim},
                {\fp_eval:n { -2 * ##1 - 2.5 } \l__ltbl_time_vunit_dim
                  - \l__ltbl_resheight_dim})}
               ]current ~ page.north ~ west
              ) rectangle
              ([shift =
                {(.4pt - \l__ltbl_reswidth_dim, .4pt + \l__ltbl_resheight_dim)}
               ]current ~ page.south ~ east);
          }
          {
            \filldraw [ fill = \l__ltbl_bg_color_tl!5, thick,
                        draw = gray, \l__ltbl_hline_type_tl ]
              ([shift = {(
                {-.4pt + \l__ltbl_reswidth_dim},
                {\fp_eval:n { -2 * ##1 - 2.5 } \l__ltbl_time_vunit_dim
                  - \l__ltbl_resheight_dim})}
               ]current ~ page.north ~ west
              ) rectangle +
              (\l__ltbl_textwidth_dim + .8pt, -2\l__ltbl_time_vunit_dim);
          }
      }
    \bool_if:NTF \l__ltbl_timelist_hidetime_bool
      {
        \dim_set:Nn \l__ltbl_timelist_shift_dim
          { -1.5\l__ltbl_time_vunit_dim }
      }
      {
```

```
      \dim_set:Nn \l__ltbl_timelist_shift_dim
        { -\l__ltbl_time_vunit_dim }
    }
  \int_step_inline:nn { \l__ltbl_time_num_int }
    {
      \node [ darkgray!80, shift =
              {(
                \l__ltbl_textwidth_dim/30,
                -2 * ##1 \l__ltbl_time_vunit_dim +
                \l__ltbl_timelist_shift_dim
              )}, font = \large \l__ltbl_timelist_numformat_tl
            ] at ([shift =
                {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                  ]current ~ page.north ~ west) {##1};
    }
  \bool_if:NF \l__ltbl_timelist_hidetime_bool
    {
      \int_step_inline:nn { \clist_count:N \l__ltbl_time_clist }
        {
          \__ltbl_get_left:eN  { \clist_item:Nn \l__ltbl_time_clist {##1} }
            \l__ltbl_tmpa_tl
          \__ltbl_get_right:eN { \clist_item:Nn \l__ltbl_time_clist {##1} }
            \l__ltbl_tmpb_tl
          \node [ gray, align = center, shift =
                  {(
                    \l__ltbl_textwidth_dim/30,
                    \fp_eval:n { -1.85 - 2 * ##1 } \l__ltbl_time_vunit_dim
                  )}, font = \l__ltbl_timelist_timeformat_tl
                ] at ([shift =
                    {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                      ]current ~ page.north ~ west)
            { \l__ltbl_tmpa_tl\\ \l__ltbl_tmpb_tl };
        }
    }
  \int_step_inline:nn { \clist_count:N \l__ltbl_week_clist }
    {
      \int_compare:nNnF {##1} = { \clist_count:N \l__ltbl_week_clist }
        {
          \node [ shift =
                  {(\fp_eval:n
                      {
                        14 * \clist_item:Nn \l__ltbl_week_accum_clist {##1}/
                        \l__ltbl_week_num_fp/15 + 1/15
                      } \l__ltbl_textwidth_dim, -2\l__ltbl_time_vunit_dim
                  )}, darkgray, font = \ttfamily
                ] at ([shift =
                      {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                      ]current ~ page.north ~ west)
            { \l__ltbl_weeklist_sep_tl };
        }
      \__ltbl_get_left:eN { \clist_item:Nn \l__ltbl_week_clist {##1} }
        \l__ltbl_tmpa_tl
      \node [ shift =
              {(\fp_eval:n
```

```
                    {
                      14(
                        \clist_item:Nn \l__ltbl_week_accum_clist {##1} -
                        \clist_item:Nn \l__ltbl_week_ratio_clist {##1}/2
                        )/\l__ltbl_week_num_fp/15 + 1/15
                    } \l__ltbl_textwidth_dim, -2\l__ltbl_time_vunit_dim
                )}, font = \large \l__ltbl_weeklist_format_tl
              ] at ([shift =
                  {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                  ]current ~ page.north ~ west) { \l__ltbl_tmpa_tl };
      }
    \draw [ gray, very ~ thick ]
      ([shift =
       {({-.4pt +\l__ltbl_reswidth_dim}, {-.4pt -\l__ltbl_resheight_dim})}
      ]current ~ page.north ~ west) rectangle
      ([shift =
       {({.4pt - \l__ltbl_reswidth_dim}, {.4pt + \l__ltbl_resheight_dim})}
      ]current ~ page.south ~ east);
  }
```

(*End of definition for* `\__ltbl_maketable:n`.)

`\l__ltbl_course_shift_dim` Store the vertical shift of the course subject in course box.

```
\dim_new:N \l__ltbl_course_shift_dim
```

(*End of definition for* `\l__ltbl_course_shift_dim`.)

`\__ltbl_course_box_aux:nn` Define the auxiliary command of `\course`.

```
\cs_new_protected_nopar:Npn \__ltbl_course_box_aux:nn #1#2
  {
    \begin{scope}
      \clip [ preaction = { draw, ultra ~ thick, \l__ltbl_course_color_tl!60 },
              preaction = { fill, \l__ltbl_course_color_tl!10 },
              rounded ~ corners = 8pt ]
        ([shift =
          {(
            \fp_eval:n
              {
                \clist_item:Nn \l__ltbl_week_accum_clist
                  { \l__ltbl_weekday_int } -
                \clist_item:Nn \l__ltbl_week_ratio_clist
                  { \l__ltbl_weekday_int }
              } \l__ltbl_week_hunit_dim + \l__ltbl_textwidth_dim/15
            + \l__ltbl_reswidth_dim + 1.2pt,
            \fp_eval:n { -.5 - 2 * #1 } \l__ltbl_time_vunit_dim
            - \l__ltbl_resheight_dim - 1.2pt
        )}]current ~ page.north ~ west) rectangle +
        (
          \clist_item:Nn \l__ltbl_week_ratio_clist
            { \l__ltbl_weekday_int } \l__ltbl_week_hunit_dim - 2.4pt,
          \fp_eval:n { 2(#1 - #2 - 1) } \l__ltbl_time_vunit_dim + 2.4pt
        );
      \fill [ \l__ltbl_course_color_tl!60 ]
        ([shift =
          {(
```

```
          \fp_eval:n
            {
              \clist_item:Nn \l__ltbl_week_accum_clist
                { \l__ltbl_weekday_int } -
              \clist_item:Nn \l__ltbl_week_ratio_clist
                { \l__ltbl_weekday_int }
            } \l__ltbl_week_hunit_dim
          + \l__ltbl_reswidth_dim + \l__ltbl_textwidth_dim/15,
          \fp_eval:n { -.5 - 2 * #1 } \l__ltbl_time_vunit_dim
          - \l__ltbl_resheight_dim
      )}]current ~ page.north ~ west) rectangle +
      (
        \clist_item:Nn \l__ltbl_week_ratio_clist
          { \l__ltbl_weekday_int } \l__ltbl_week_hunit_dim,
        -\l__ltbl_time_vunit_dim/2
      );
  \end{scope}
  \int_compare:nNnTF {#1} = {#2}
    {
      \bool_lazy_and:nnTF
        { \tl_if_empty_p:N \l__ltbl_course_location_tl   }
        { \tl_if_empty_p:N \l__ltbl_course_lecture_tl    }
        { \tl_set:Nn \l__ltbl_course_anchor_tl {       } }
        { \tl_set:Nn \l__ltbl_course_anchor_tl { above } }
      \node
        [ \l__ltbl_course_anchor_tl, \l__ltbl_course_color_tl!60, shift =
          {(
            \fp_eval:n
              {
                \clist_item:Nn \l__ltbl_week_accum_clist
                  { \l__ltbl_weekday_int } -
                \clist_item:Nn \l__ltbl_week_ratio_clist
                  { \l__ltbl_weekday_int }/2
              } \l__ltbl_week_hunit_dim + \l__ltbl_textwidth_dim/15,
            \fp_eval:n { -1.75 - #1 - #2 } \l__ltbl_time_vunit_dim
          )}, align = center, font = \bfseries
        ] at ([shift =
            {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
              ]current ~ page.north ~ west) { \l__ltbl_course_subject_tl };
      \bool_lazy_or:nnTF
        { \tl_if_empty_p:N \l__ltbl_course_location_tl }
        { \tl_if_empty_p:N \l__ltbl_course_lecture_tl  }
        { \tl_set:Nn \l__ltbl_s@course_sep_tl {    }   }
        { \tl_set:Nn \l__ltbl_s@course_sep_tl { ,~ }   }
      \node
        [ shift =
          {(
            \fp_eval:n
              {
                \clist_item:Nn \l__ltbl_week_accum_clist
                  { \l__ltbl_weekday_int } -
                \clist_item:Nn \l__ltbl_week_ratio_clist
                  { \l__ltbl_weekday_int }/2
              } \l__ltbl_week_hunit_dim + \l__ltbl_textwidth_dim/15,
```

```
            \fp_eval:n { -1.75 - #1 - #2 } \l__ltbl_time_vunit_dim
          )}, below, \l__ltbl_course_color_tl!60, align = center
        ] at ([shift =
            {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
              ]current ~ page.north ~ west)
        {
          \l__ltbl_course_location_tl
          \l__ltbl_s@course_sep_tl
          \l__ltbl_course_lecture_tl
        };
    }
    {
      \bool_lazy_and:nnTF
        { \tl_if_empty_p:N \l__ltbl_course_location_tl }
        { \tl_if_empty_p:N \l__ltbl_course_lecture_tl  }
        {
          \tl_set:Nn \l__ltbl_course_anchor_tl { }
          \dim_set:Nn \l__ltbl_course_shift_dim { 0pt }
        }
        {
          \tl_set:Nn \l__ltbl_course_anchor_tl { above }
          \dim_set:Nn \l__ltbl_course_shift_dim { \l__ltbl_time_vunit_dim/8 }
        }
      \node
        [ \l__ltbl_course_color_tl!60, align = center, shift =
          {(
            \fp_eval:n
              {
                \clist_item:Nn \l__ltbl_week_accum_clist
                  { \l__ltbl_weekday_int } -
                \clist_item:Nn \l__ltbl_week_ratio_clist
                  { \l__ltbl_weekday_int }/2
              } \l__ltbl_week_hunit_dim + \l__ltbl_textwidth_dim/15,
            \fp_eval:n { -1.5 - #1 - #2 } \l__ltbl_time_vunit_dim +
            \l__ltbl_course_shift_dim
          )}, font = \large \bfseries, \l__ltbl_course_anchor_tl
        ] at ([shift =
            {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
              ]current ~ page.north ~ west) { \l__ltbl_course_subject_tl };
      \bool_lazy_or:nnTF
        { \tl_if_empty_p:N \l__ltbl_course_location_tl }
        { \tl_if_empty_p:N \l__ltbl_course_lecture_tl  }
        { \tl_set:Nn \l__ltbl_course_sep_tl {    }     }
        { \tl_set:Nn \l__ltbl_course_sep_tl { \\ }     }
      \node
        [ shift =
          {(
            \fp_eval:n
              {
                \clist_item:Nn \l__ltbl_week_accum_clist
                  { \l__ltbl_weekday_int } -
                \clist_item:Nn \l__ltbl_week_ratio_clist
                  { \l__ltbl_weekday_int }/2
              } \l__ltbl_week_hunit_dim + \l__ltbl_textwidth_dim/15,
```

```
              \fp_eval:n { -1.625 - #1 - #2 } \l__ltbl_time_vunit_dim
            )}, below, \l__ltbl_course_color_tl!60, align = center
          ] at ([shift =
              {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                ]current ~ page.north ~ west)
          {
            \l__ltbl_course_location_tl
            \l__ltbl_course_sep_tl
            \l__ltbl_course_lecture_tl
          };
        \node
          [ shift =
            {(
              \clist_item:Nn \l__ltbl_week_accum_clist
                { \l__ltbl_weekday_int } \l__ltbl_week_hunit_dim +
              \l__ltbl_textwidth_dim/15,
              \fp_eval:n { -2.5 - 2 * #2 } \l__ltbl_time_vunit_dim
            )}, above ~ left, \l__ltbl_course_color_tl!60, outer ~ sep = .5ex
          ] at ([shift =
              {(\l__ltbl_reswidth_dim, -\l__ltbl_resheight_dim)}
                ]current ~ page.north ~ west) { \l__ltbl_course_comment_tl };
      }
  }
```

(*End of definition for* `\__ltbl_course_box_aux:nn`.)

```
\file_input_stop:
```

End the optionlist package for l3docstrip.

⟨/package⟩

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